



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

AI-BASED INTERVIEW SIMULATOR WITH INTELLIGENT FEEDBACK

The domain of the Project: Artificial Intelligence

COURSE NAME

**Project-Based Internship in Artificial Intelligence & Machine
Learning**

Team Mentors (and their designation): Gaurav Patel -Data Engineer

Team Members:

Ms. Muthyala Deepika Reddy

Period of the project

May 2026 to June 2026



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Declaration

The project titled “AI-Based Interview Simulator With Intelligent Feedback” has been mentored by Gaurav Patel, organised by SURE Trust, from December to June 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. Muthyala Deepika Reddy
Signature M.Deepika

Mentor's Name Gaurav Patel
Designation—Company Name
Data Engineer

Mentor's Name Gaurav Patel
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Executive Summary

AI-Powered Interview Preparation System is developed to help students and job seekers improve their interview skills through an intelligent and interactive platform. The main objective of this project is to provide a realistic interview practice environment where users can answer technical and HR interview questions and receive instant AI-based feedback. The system aims to improve technical knowledge, communication skills, and overall interview confidence.

The project is developed using Python, Streamlit, Groq LLM, gTTS, and Speech Recognition. It generates interview questions based on the selected domain and difficulty level. The AI evaluates each answer and provides scores, strengths, weaknesses, improvement suggestions, and an ideal answer. The system also includes voice-based features, allowing users to listen to interview questions and answer using voice, making the interview experience more interactive and practical.

The key findings of this project show that AI can provide personalized feedback and help users identify their mistakes, improve their responses, and prepare more effectively for real interviews. The additional features, such as Review My Interview and Learn Interview Questions, further support users by analyzing real interview answers and explaining important interview concepts.

It is recommended that this system be enhanced in the future by adding features such as resume-based interview question generation, multilingual support, video interview analysis, emotion detection, and detailed performance tracking. These improvements will make the platform more intelligent, user-friendly, and beneficial for students, fresh graduates, and job seekers preparing for different types of interviews.



Introduction

Background and Context of the Project

Interviews are an important part of the recruitment process for students and job seekers. Many candidates have good technical knowledge but find it difficult to perform well in interviews because of a lack of confidence, practice, and proper guidance. Traditional interview preparation methods often do not provide personalized feedback or a realistic interview experience. With the growth of Artificial Intelligence, it is now possible to create intelligent systems that help users practice interviews and receive instant feedback. The **AI-Powered Interview Preparation System** is developed to provide a smart, interactive, and user-friendly platform for interview preparation.

Problem Statement

Many students and fresh graduates face difficulties in preparing for technical and HR interviews due to limited access to mock interviews and expert feedback. Existing platforms may not provide personalized evaluation or detailed suggestions for improvement. The main goal of this project is to develop an AI-based interview simulator that generates interview questions, evaluates user answers, provides detailed feedback, and helps users improve their interview performance and confidence before attending real interviews.

Scope and Limitations of the Project

The system supports multiple interview domains, including Python, Machine Learning, Data Science, AI/ML, Web Development, and HR interviews. Users can select the interview domain, difficulty level, and number of questions. The application generates interview questions, evaluates answers, provides scores, strengths, weaknesses, improvement suggestions, and ideal answers. It also includes voice-based features, interview review, and concept learning modes.

The current version has some limitations. The quality of the evaluation depends on the AI model and the user's responses. It requires an internet connection to access AI services, and it currently supports only the available interview domains. Features such as resume-based interviews, video interview analysis, emotion detection, and multilingual support are not included in this version.

Innovation Component in the Project

The main innovation of this project is the integration of Artificial Intelligence with interview preparation. Unlike traditional mock interview systems, this application generates intelligent interview questions and provides personalized feedback using a Large Language Model. It also combines text-to-speech and speech-to-text technologies to support voice-based interviews, creating a more realistic interview experience. The additional features, such as Review My Interview and Learn Interview Questions, make the system more interactive and useful for continuous learning and interview preparation.



Project Objectives

Objectives of the Project

- The main objective of the AI-Powered Interview Preparation System is to help students and job seekers prepare for technical and HR interviews through an intelligent and interactive platform. The project aims to provide a realistic interview experience by generating AI-based interview questions and evaluating user responses with personalized feedback.
- The specific objectives of the project are:
- To develop an AI-based interview preparation platform using Python and Streamlit.
- To generate domain-specific interview questions based on the user's selected domain and difficulty level.
- To evaluate user answers and provide technical, communication, and completeness scores.
- To identify the strengths and weaknesses of the user's answers and provide improvement suggestions.
- To support text-based and voice-based interview practice using text-to-speech and speech-to-text technologies.
- To provide Review My Interview and Learn Interview Questions features to help users analyze answers and learn interview concepts.
- To improve users' confidence, communication skills, and interview performance before attending real job interviews.

Expected Outcomes and Deliverables

- The expected outcome of this project is an intelligent interview preparation system that helps users practice interviews and receive instant AI-based feedback. The application provides a realistic interview environment where users can improve their knowledge, communication skills, and confidence.
- The final deliverables of the project include:
- A fully functional AI-Powered Interview Preparation System.
- AI-generated interview questions with intelligent answer evaluation and detailed feedback.
- Voice-based interview support using text-to-speech and speech-to-text.
- Interview review, learning modules, and a final performance report.
- Source code, project documentation, and deployment on Streamlit.



Methodology and Results

Methods / Technology Used

The project was developed using **Python** as the programming language and **Streamlit** for building the user interface. **Groq LLM** was used to generate interview questions, evaluate user answers, and provide feedback. **gTTS (Google Text-to-Speech)** was used to read interview questions aloud, while **Speech-to-Text** technology was used to convert voice responses into text for evaluation.

Tools / Software Used

- Python
- Streamlit
- Groq LLM
- gTTS (Google Text-to-Speech)
- Speech Recognition
- Streamlit Mic Recorder
- Visual Studio Code
- GitHub

Data Collection Approach

The project does not use a fixed dataset. Interview questions and feedback are generated dynamically using the Groq Large Language Model based on the selected interview domain and difficulty level. User responses are collected through text or voice input and evaluated in real time.

Project Architecture

The system begins by allowing users to select the interview mode, domain, difficulty level, and number of questions. The Groq LLM generates interview questions based on these selections. Users answer the questions using text or voice input. Voice responses are converted into text using Speech-to-Text technology. The AI then evaluates the answers and provides scores, strengths, weaknesses, improvement suggestions, and ideal answers. Finally, the application generates a performance report summarizing the user's interview results.



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Final Project Working Screenshots

- Home Page (Interview Settings and Mode Selection)

The screenshot shows the 'AI Based Interview Simulator with Intelligent Feedback' interface. On the left, there's a sidebar with 'Interview Settings' including 'Select Domain' (Python), 'Select Difficulty' (Beginner), and a 'Number of Questions' slider. A 'Report' button is at the bottom of the sidebar. The main area has a title 'AI Based Interview Simulator with Intelligent Feedback' and a subtitle 'Practice interviews, analyze real interview answers, and learn interview concepts using AI.' Below this is a 'Choose Mode' section with three radio buttons: 'Mock Interview' (selected), 'Review My Interviews', and 'Learn Interview Questions'. At the bottom, a 'Configuration' section displays four boxes: 'Mode: Mock Interview', 'Domain: Python', 'Difficulty: Beginner', and 'Questions: 5'. A 'Deploy' link is in the top right corner.

- Mock Interview Page

The screenshot shows the 'AI Interview Session' interface. It features a title 'AI Interview Session' and a subtitle 'Question 1 of 5'. Below this is a text box containing the question: 'Write a Python function to reverse a given string, for example, if the input is "hello", the output should be "olleh".' A progress bar shows '0:00 / 0:00'. Below the question is a large text area for 'Your Answer' with the placeholder 'Type your answer here...'. At the bottom, there's a 'Voice Answer' section with a microphone icon and a 'Start Recording' button. A 'Deploy' link is in the top right corner.



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- Voice Recording and Speech-to-Text Feature

Deploy / |

AI Interview Session

Question 1 of 5

What is the difference between supervised and unsupervised learning in machine learning, and provide an example of each?

▶ 0:00 / 0:08 🔊 ⋮

Your Answer

Type your answer here...

Voice Answer

Stop Recording

Voice Recorded Successfully

- AI Answer Evaluation Page

Deploy / |

Answer Evaluated Successfully!

Technical Score	Communication Score	Completeness Score
20	15	20

Strengths

- ✔ Correctly identifies the basic difference between supervised and unsupervised learning
- ✔ Mentions the use of labelled and unlabelled data

Weaknesses

- ✘ Lacks specific examples for each type of learning
- ✘ Does not fully articulate the goals or applications of supervised and unsupervised learning
- ✘ Sentence structure is incomplete and slightly confusing

Improvement Tips

- 📌 Provide concrete examples for both supervised and unsupervised learning to illustrate the concepts
- 📌 Clearly define the objectives and common applications of each learning type
- 📌 Improve sentence structure for better clarity and readability



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- Review My Interview Page

Deploy

AI Interview Session

 **Review My Interview**

Interview Question

Your Answer

Analyze Answer

- Learn Interview Questions Page

Deploy

AI Interview Session

 **Learn Interview Questions**

Enter Any Topic

python

Explain

Here's an overview of the topic: "Python".

1. Definition: Python is a high-level, interpreted programming language that is widely used for various purposes such as web development, scientific computing, data analysis, artificial intelligence, and more. It was created in the late 1980s by Guido van Rossum and was first released in 1991. Python is known for its simplicity, readability, and ease of use, making it a popular choice among beginners and experienced programmers alike.

2. Important Concepts: Some important concepts in Python include:

- **Variables and Data Types:** Python has various data types such as integers, floats, strings, lists, tuples, and dictionaries.
- **Control Structures:** Python has control structures such as if-else statements, for loops, and while loops.
- **Functions:** Python functions are reusable blocks of code that take arguments and return values.
- **Modules and Packages:** Python has a vast collection of libraries and modules that can be imported to perform various tasks.
- **Object-Oriented Programming (OOP):** Python supports OOP concepts such as classes, objects, inheritance, and polymorphism.
- **File Input/Output:** Python provides functions for reading and writing files, including text files, CSV files, and JSON files.

3. Interview Questions: Here are some common Python interview questions:



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- Final Report Page



Project GitHub Link

GitHub Repository:

<https://github.com/sure-trust/MUTHYALA-DEEPIKA-REDDY-g37-ai-ml>



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Learning and Reflection

New Learnings

Through this project, I gained practical knowledge of Python, Streamlit, Groq LLM integration, Speech-to-Text, Text-to-Speech, and GitHub. I also learned how to design an AI-based application, solve technical challenges, and improve my problem-solving, project planning, and software development skills.

Overall Experience

Developing this project was a valuable learning experience for me. During development, I faced challenges while integrating AI features, implementing speech recognition, and fixing technical issues, but solving them helped me improve my programming and problem-solving skills. This project also gave me confidence in building a complete AI application from design to deployment and motivated me to continue learning advanced AI technologies.



Conclusion and Future Scope

Conclusion

The AI-Powered Interview Simulation System was successfully developed to help students and job seekers improve their interview skills. The project achieved its objectives by generating AI-based interview questions, evaluating user responses, providing personalized feedback, and supporting both text and voice-based interviews. Overall, the system offers an effective and user-friendly platform for interview preparation.

Future Scope

The project can be enhanced by adding support for more technical domains and programming languages. Future versions can include video interview analysis, resume-based question generation, multilingual support, and detailed performance tracking. These improvements will make the system more intelligent and useful for a wider range of users.